GH3036 Customer Project Parameter Adjustment Guide

Goodix Confidential For lierda_hd@lierda.com Only

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Preface

Purpose

This document is intended for users of GH3036 chipset who want to configure parameters to adapt to different applications.

Audience

This document is intended for:

- Software developers
- Systems and product engineers
- Field application engineers

Release Notes

This document is the first release.

Revision History

Version	Date	Description
1.0	2024-10-14	Initial release.
2.0	2025-03-05	Update for new registers
2.1	2026-01-05	1. Added VLED Low-Voltage Power Supply Configuration 2. Updated DRE register notice 3. Adde HR DRE configuration

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1 Introduction

This document is intended for users of GH3036 who want to configure parameters to adapt to different applications by using GH(M)3036 EVK tool. Before reading this document, it is strongly recommended that you read the relevant chip datasheet and “GHealth EVK PC Tool User Manual”.

Chapter 1 introduces the purpose of the document. Chapter 2 descripts the process of configuration. Chapter 3 illustrates the process of different configuration items. And chapter 4 gives some examples of how users modify configurations.

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2 Configuration process

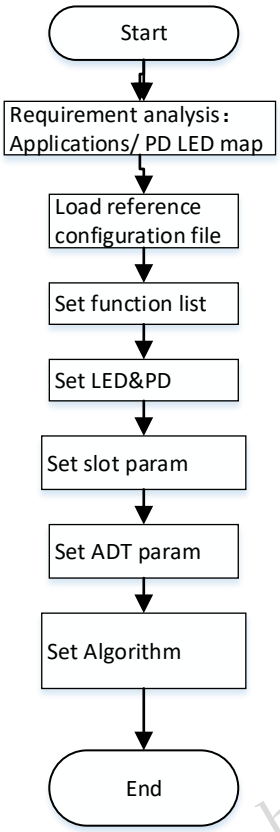

Figure 2-1 Configuration process

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The configuration process as shown in the figure above:

1. Requirement analysis. At the beginning of the configuration process, it is important to identify the specific functions that need to be configured, such as HR, SPO2 and gathering the information of PD/LED map.
2. It is difficult for a user to configure all parameters from zero, so the reference configuration files are needed. At this step, load the reference configuration file which include the needed functions.
3. According to the actual situation, select the corresponding functions.
4. According to the actual optical path, set LED and PD map.
5. Set AGC parameters if needed.
6. Set ADT parameters and logic if needed.
7. Set Algorithm if needed.

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3 Configuration Description

3.1 Select Function

Choose the application functions. Select the functions if these functions need to run at the same time.

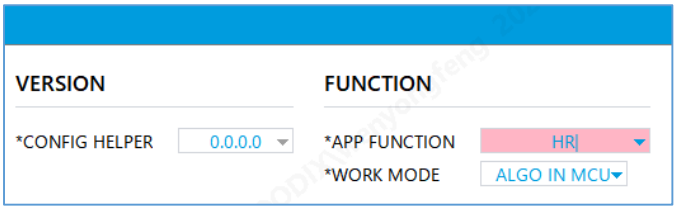

Figure 3-1 Function Items

The description of each items are shown below.

Table 3-1 Function Items Description

Item	Description	Notice
APP FUNCTION	To use the raw data sampled by the chip for algorithm processing, it is needed to select APP FUNCTIONS.	If only focusing on raw data, this item can be left unselected.
WORK MODE	0: EVK. Only generate raw data to evaluate chip performance. 1: ALGO IN APP. The algorithm of APP FUNCTION will run in PC or Mobile phone. 2: ALGO IN MCU. The algorithm of APP FUNCTION will run in MCU.	WORK MODE can only be configured as 1 or 2 if **APP FUNCTION is non-zero**.

3.2 Time Slot

3.2.1 LED Configuration

Step 1. Get hardware circuit. Example as follows:

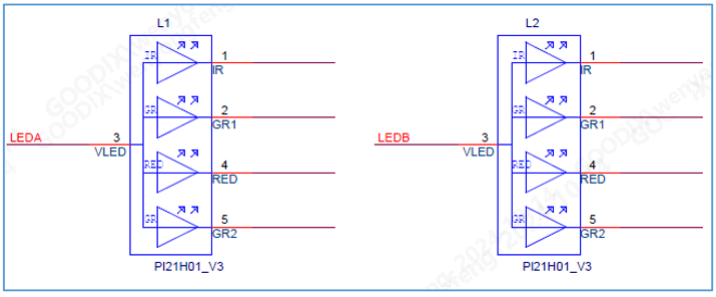

Figure 3-2 hardware circuit

Step 2. Assign a number to each LED.

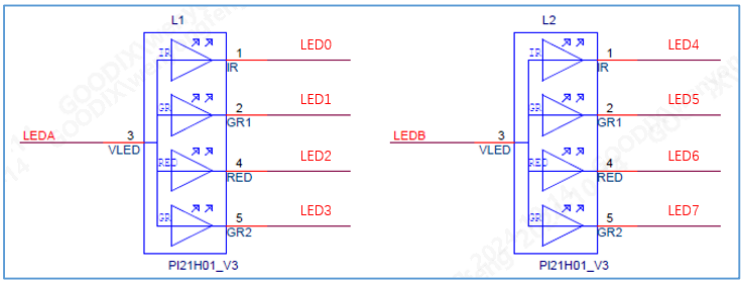

Figure 3-3 LED numbers

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Step 3. Configure the LEDs according to hardware circuit and the numbers. Please refer to datasheet of the chipset to get more information about LED drivers.

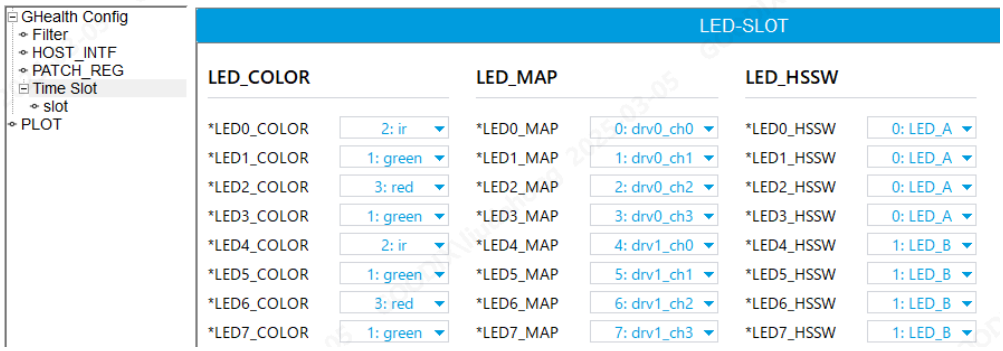

Figure 3-4 LED Configuration

Table 3-2 LED Configuration

Item	Description	Notice
LEDx_COLOR	Set the color of LEDx	Depends on the actual hardware circuit
LEDx_MAP	Set the driver pins of LEDx	Depends on the actual hardware circuit
LEDx_HSSW	Set the hide-side switch of LEDx	Depends on the actual hardware circuit
LEDDRVx_FS	Set the max current of LED Driverx	-

3.2.2 PD Configuration

Set PD Mode.

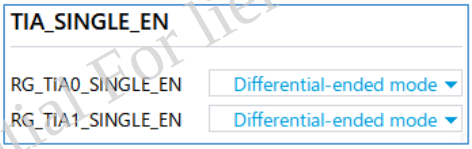

Figure 3-5 PD mode

3.2.3 SLOT_CFG_NUM

The "slot" refers to the sampling time slot, while "slot cfg" is a set of registers used to control the sampling. Setting the "slot index" means setting the use of a specific group of registers for that sampling time slot. As shown in the following figure, use "ppg_cfg_0" for "Slot 0" and use "cap_cfg_13" for "Slot 1".

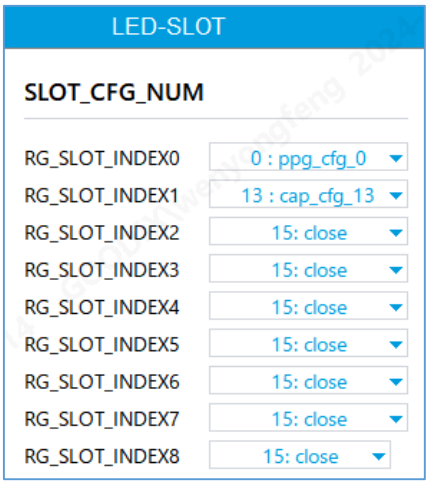

Figure 3-6 slot cfg num

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3.3 PPG Slot Parameter

3.3.1 LED & PD Select

3.3.1.1 Slot LED select

Select which LED works at sample slots (At page Slot => SLOT CFG xx(PPG)).

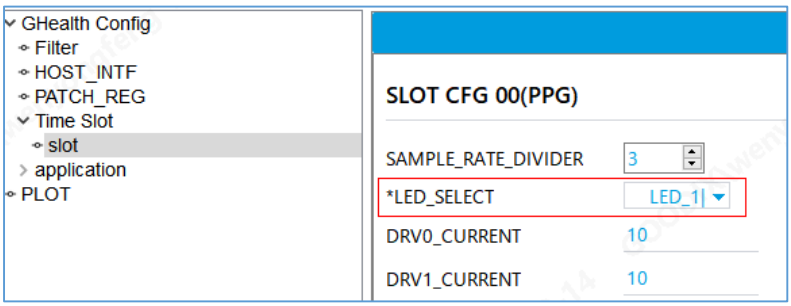

Figure 3-7 Slot LED select

3.3.1.2 Slot PD select

Select which PD works at sample slots (At page Slot => SLOT CFG xx(PPG)).

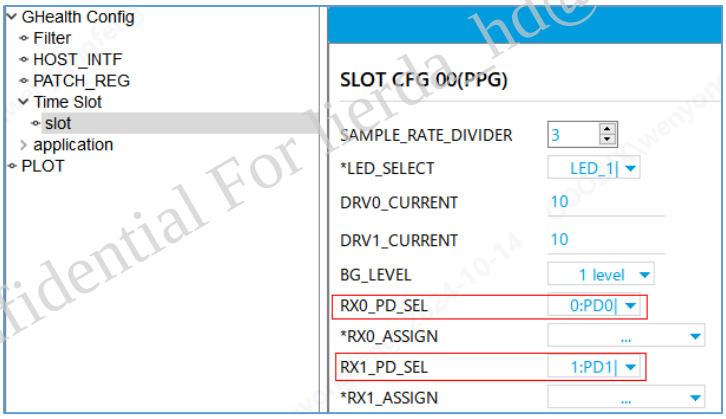

Figure 3-8 Slot PD select

Note: If there are idle RX and idle PD, the tool will automatically map the idle PD to the idle RX.

3.3.2 RX_ASSIGN

If the APP FUNCTION selection (illustration at chapter 3.1) is not empty, RX_ASSIGN should be configured. RX_ASSIGN means to assign the raw data of the correspond channel to different application functions. Example as follows (At page Slot => SLOT CFG xx(PPG)), the raw data of RX0 at slot 0 is assigned to HR application.

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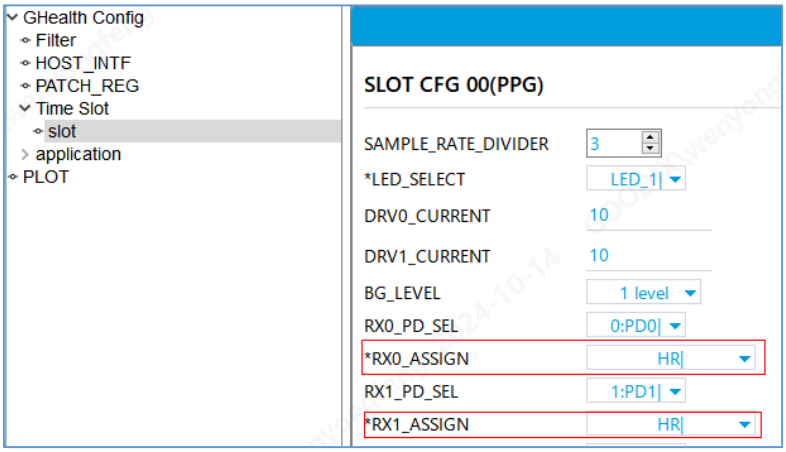

Figure 3-9 RX_ASSIGN

3.3.3 Slot Sample Rate

The sample rates of the slots depend on 2 divider parameters (At page HOST_INTF => SAMPLE and slot).

Slot sample Rate = 32000/((1+FASTEST_SAMPLE_RATE_DIVIDER) *(1+SAMPLE_RATE_DIVIVER))

Example as follows, the sample rate of slot 0 is set to 25Hz.

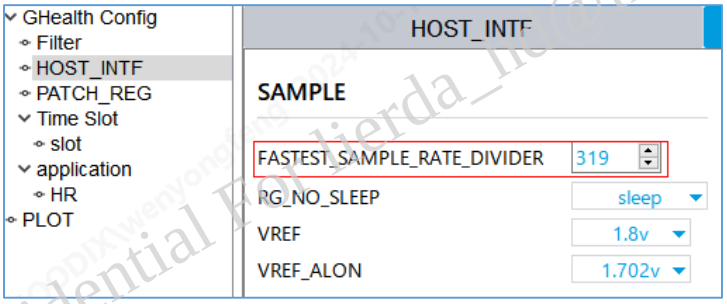

Figure 3-10 Set fastest sample rate

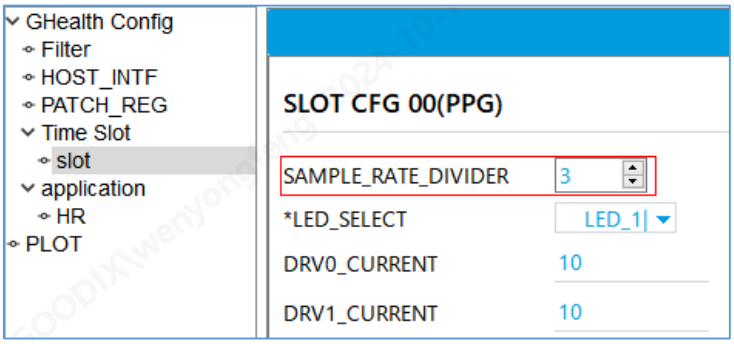

Figure 3-11 Set slot sample rate

3.3.4 Other Slot Parameters

The other slot parameter is recommended to align with reference configuration.

Table 3-3 Slot Parameters

Item	Description	Notice
DRVx_CURRENT	The current of DRVx at this slot. Unit: mA.	Cannot exceed the maximum value (LEDDRVx_FS at Table 3-2)

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Item	Description	Notice
BG_LEVEL	BG rejection level : 0,1,2,4.	-
RXx_TIA	TIA gain of RXx at this slot.	
RXx_TIA_CF	TIA capacitance of RXx at this slot.	
BG_CANCEL_ENABLE	Enable/Disable BG cancel.	
BG_LV0_POW_ENABLE	Enable/Disable pull LED power high when BG LEVEL is 0 level.	
BG_CANCEL_SKIP_ENABLE	Enable/Disable BG cancel skip.	
RXx_BG_CANCEL_RANGE	The max current of BG cancel.	
DC_CANCEL_ENABLE	Enable/Disable DC cancel.	
RXx_DC_CANCEL_RANGE	The max current of DC cancel	
RXx_DC_CANCEL	DC cancel code.	
DRE_ENABLE	Enable/Disable DRE.	DC Cancel must be enabled when using DRE.
DRE_FIFO_OUTPUT_EN	Enable DRE DC info data output to FIFO.	DC info data is an extra data set containing data before DC cancel compensation.
DRE_SCALE	DRE dynamic range extension ratio: 2x ~ 256x.	The data in FIFO is right shifted by DRE_SCALE bit.
DRE_CANCEL_CODE	DRE DC cancel compensation decreases dre_cancel_code*16 for each frame until to 0	Recommended to use only in HR.
ADC_INT	ADC Integration time.	
FIFO_CTRL	Controls the type of data written into fifo.	
TIA_SET_TIME	TIA set state duration selection.	
SLOT_INNER_DELAY	Anti-crosstalk delay between Bg, mix, and dre sampling.	
SLOT_POST_DELAY	Anti-crosstalk delay after slot sampling.	
AGC_EN	Soft agc slot adjust enable.	
AGC_SPO2_RED_EN	Enable SPO2 red light soft agc.	
AGC_SPO2_IR_EN	Enable SPO2 IR light soft agc.	
AGC_SCR	The main dimming channel selection of soft agc.	
AGC_DRV	Dimming LED DRV selection of soft agc.	
AGC_DRV_UP_LIMIT	Up limit for soft agc.	
AGC_DRV_DOWN_LIMIT	Down limit for soft agc.	

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3.4 CAP Slot Parameter

Select “13:cap_cfg_13” in the SLOT_CFG_NUM option.

3.4.1 Slot Sample Rate

The sample rates of the slots depend on 2 divider parameters (At page HOST_INTF => SAMPLE and slot).

Slot sample Rate = 32000/((1+FASTEST_SAMPLE_RATE_DIVIDER) *(1+SAMPLE_RATE_DIVIVER))

Example as follows, the sample rate of slot 0 is set to 25Hz.

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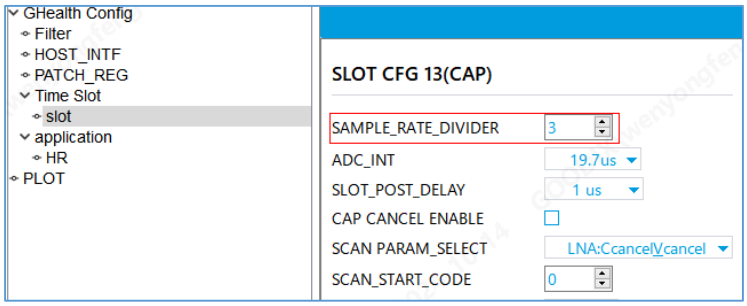

Figure 3-12 Set slot sample rate

3.4.2 RX_ASSIGN

If the APP FUNCTION selection (illustration at chapter 3.1) is not empty, RX_ASSIGN should be configured. RX_ASSIGN means to assign the raw data of the correspond channel to different application functions. Example as follows (At page Slot => SLOT CFG 13(CAP)), the raw data of cap at slot 0 is assigned to HR application.

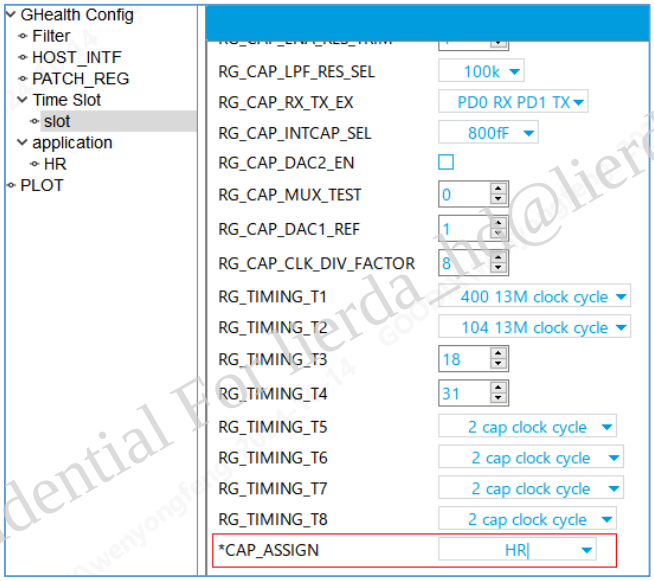

Figure 3-13 RX_ASSIGN

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3.4.3 Other Slot Parameters

The other slot parameter is recommended to align with reference configuration.

Table 3-4 Slot Parameters

Item	Description	Notice
ADC_INT	ADC Integration time.	Cannot exceed the maximum value (LEDDRVx_FS at Table 3-2)
SLOT_POST_DELAY	Anti-crosstalk delay after slot sampling.	-
CAP_CANCEL_ENABLE	Enable/Disable Cap cancel.	
SCAN PARAM_SELECT	Cap cancel scan parameter selection.	
SCAN_START_CODE	Cap cancel scan start code.	High 2bit : Ccancel code,low 8bit : Vcancel code
SCAN_STEP	Cap cancel scan step.	-
SCAN_STOP_CODE	Cap cancel scan stop code.	High 2bit : Ccancel code,low 8bit : Vcancel code
RG_LNA_CANCEL_CODE	In non-cap_cancel scanning mode, control DA_CAP_DAC1.	-

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Item	Description	Notice
RG_SCI_CANCEL_CODE	In non-cap_cancel scanning mode, control DA_CAP_DAC2.	
RG_CAP_SLP_SEL	Buffer pow control selection of DAC2 &amp	
RG_CAP_LNA_RES_SEL	Trim LNA inner res.	
RG_CAP_LPF_RES_SEL	LPF res selection.	
RG_CAP_RX_RX_EX	RX TX exchange.	
RG_CAP_INTCAP_SEL	INT cap selection.	
RG_CAP_DAC2_EN	Enable/Disable Cap DAC2.	
RG_CAP_MUX_TEST	Cap mux test.	
RG_CAP_DAC1_REF	Cap DAC1 ref.	
RG_CAP_CLK_DIV_FACTOR	Cap_ck divider: {8:2:64},13M divided clock.	
RG_TIMING_T1	Under CAP timing, T1 range.	
RG_TIMING_T2	Under CAP timing, T2 range.	
RG_TIMING_T3	Under CAP timing, T3 range.	T3 range: {4:2:128} cap_ck cycles, it is necessary to ensure that T3 time is greater than T1+T2 time. If this register is configured as 6'd63, it corresponds to 1 cap_ck, which is an illegal configuration.
RG_TIMING_T4	Under CAP timing, T4 range.	{2:2:1024} cap_ck cycles.
RG_TIMING_T5	Under CAP timing, T5 range.	-
RG_TIMING_T6	Under CAP timing, T6 range.	
RG_TIMING_T7	Under CAP timing, T7 range.	there is no constraint relationship between T6 and T7. T6 can be greater than, equal to, or less than T7.
RG_TIMING_T8	Under CAP timing, T8 range.	there is no constraint relationship between T5 and T8. T5 can be greater than, equal to, or less than T8.

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3.5 ADT

3.5.1 Add ADT function

Step 1. Select ADT in function List

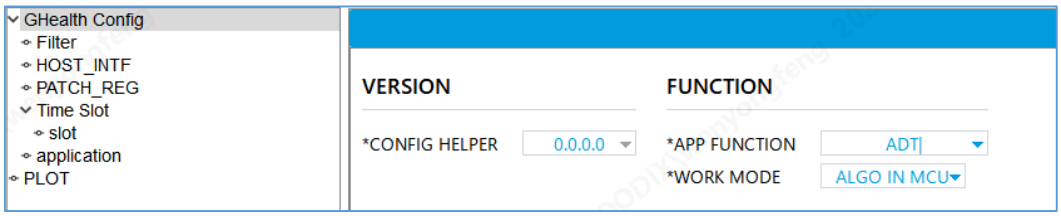

Figure 3-14 Add ADT Function

Step 2. Set ADT Slot parameters (At page Slot). Notice to set the ADT slot parameters the same as the reference configuration.

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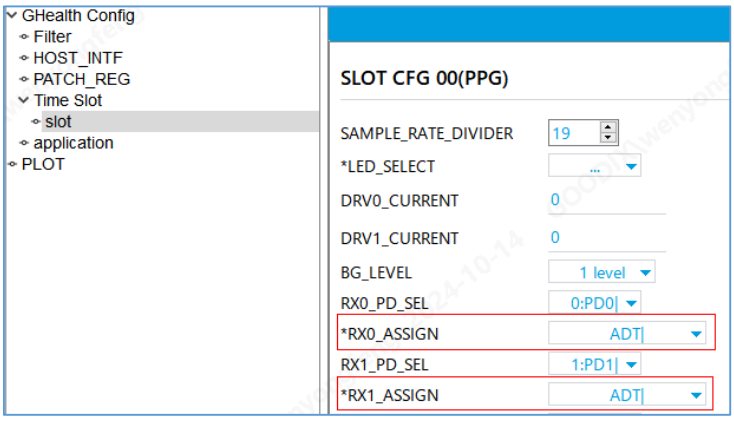

Figure 3-15 ADT Slot parameters

3.6 HOST_INTF

3.6.1 HOST_INTF

Table 3-5 HOST_INTF

Item	Description	Notice
RG_INT_AFTER_FRAME	FIFO full interrupt will be generated after sampling ends.	-
RG_INT_PWRUP	Any interrupt will power up the system first, and then generate an interrupt.	
INT_TYPE	INT type.	
INT_WIDTH	INT pulse width.	
INT_COLD_TIME	INT cold time.	
INT_TIMER_SEL	INT timer set.	
INT_TIMER_MODE	INT timer mode.	
FIFO_WATER_LINE	FIFO water line.	Must lager than 0
RG_FIFO_READ_INT_TIMER	time interval of checking fifo over waterline pulse.	-

Table 3-6 INT_ENALBE

Item	Description	Notice
XXXX_EN	Enable INT.	-

Table 3-7 SYNC

Item	Description	Notice
RG_SYNC_MODE	Sync mode.	-
RG_SYNCIN_POLAR_SEL	Sync in polar.	
RG_SYNCIN_PIN_SEL	Sync in/out pad selection.	
RG_SYNC_OUT_MODE	Sync out mode .	
RG_SLOT_SYNC_OUT_EN	Sync out polar.	
RG_FRAME_SYNC_T1_SET	Frame Sync T1.	
RG_FRAME_SYNC_T2_SET	Frame Sync T2.	

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3.6.2 SAMPLE

Table 3-8 SAMPLE

Item	Description	Notice
RG_FASTEST_SAMPLE_RATE	Fastest_sample_rate=32000/(setting_val+1).	-
RG_NOSLEEP	Enable no sleep mode.	
VREF	Vref.	
VREF_ALON	Vref alon.	

Table 3-9 SAMPLE

Item	Description	Notice
RG_DRE_DC_THR	DRE threshold. Upper threshold: rg_dre_dc_thr * 2048; lower threshold: -rg_dre_dc_thr * 2048	DRE will be triggered when data is larger than upper threshold or lower than lower threshold.
RG_DRE_DC_THR_CNT	Anti-shake points.	DRE will be triggered when RG_DRE_DC_THR_CNT points exceed the DRE threshold.
DRE SAT HANDLE ENABLE	Enable/Disable DRE saturation processing.	DRE stops when the data is saturated and led current can be changed by software to handle saturated; DRE resumes when data is not saturated.
RG_DRE_CODE_MODE	DRE compensation update mode when saturated: 0: use mixdata and dredata to update 1: use mixdata to update	If there are many scenarios of data saturation, it is recommended to set mode=1 to get accurate data.
RG_PPG_SA_THD	Ppgmixdata saturation threshold.	rg_ppg_sa_thd * 2048

Table 3-10 MODE

Item	Description	Notice
DC MODE ENABLE	Enable/Disable DC mode.	-
RG_BG_CANCEL_MODE	BG cancel mode selection.	
RG_BG_CANCEL_CODE	BG cancel code when manual mode.	
RG_VCM_PULSEMODE	Enable/Disable pulse mode	

Table 3-11 ANA_CONFIG

Item	Description	Notice
RG_ADC_XP_EN	Enable/Disable chopping.	-
RG_ADC_XP_SL	Chopping frequency selection.	

3.6.3 SOFT_AGC

It is recommended to align with the reference configuration.

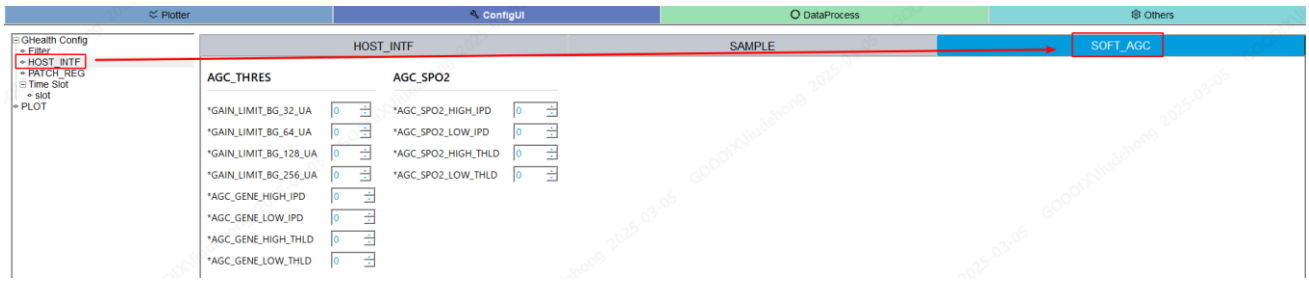

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Figure 3-16 AGC Parameters

Table 3-12 AGC_THRES

Item	Description	Notice
GAIN_LIMIT_BG_32_UA	Upper limit of gain for 32ua.	-
GAIN_LIMIT_BG_64_UA	Upper limit of gain for 64ua.	
GAIN_LIMIT_BG_128_UA	Upper limit of gain for 128ua.	
GAIN_LIMIT_BG_256_UA	Upper limit of gain for 256ua.	
AGC_GENE_HIGH_IPD	Ipd target interval upper value (unit: nA) except spo2.	
AGC_GENE_LOW_IPD	Ipd target interval lower value (unit: nA) except spo2.	
AGC_GENE_HIGH_THLD	The upper 16 bits of the upper threshold of raw data for functions except spo2.	
AGC_GENE_LOW_ THLD	The upper 16 bits of the lower threshold of raw data for functions except spo2.	

Table 3-13 AGC_SPO2

Item	Description	Notice
AGC_ SPO2_HIGH_IPD	Ipd target interval upper value (unit: nA) for spo2.	-
AGC_ SPO2_LOW_IPD	Ipd target interval lower value (unit: nA) for spo2.	
AGC_ SPO2_HIGH_THLD	The upper 16 bits of the upper threshold of raw data for spo2.	
AGC_ SPO2_LOW_ THLD	The upper 16 bits of the lower threshold of raw data for spo2.	

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3.7 Application

If the APP FUNCTION selection (illustration at chapter 3.1) is not empty, then the corresponding parameters should be configured.

3.7.1 GSENSOR_CTRL

Set g-sensor control.

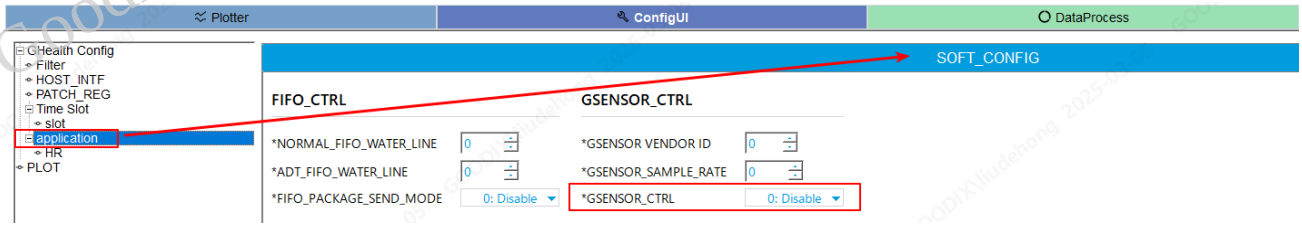

Figure 3-17 Enable/disable g-sensor

3.8 Algorithm Parameters

3.8.1 Sample

Set the fs parameter for each function at “xx_SAMPLE” column.

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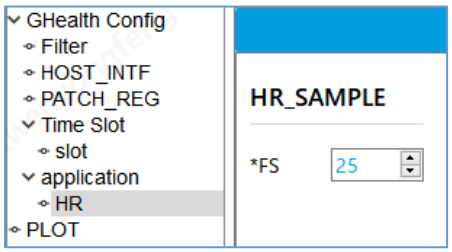

Figure 3-18 HR Param

3.8.2 Algorithm Channel

Algo_xxx_chnls are set to confirm the Algorithms (usually HR, SPO2) channels. Example as follows, the LEDs at slot 0~3 are respectively GREEN, RED and IR. And the RDx_ASSIGN are set to HR, SPO2 which marked on the figure as “HR_0”, “SPO2_0” and etc.

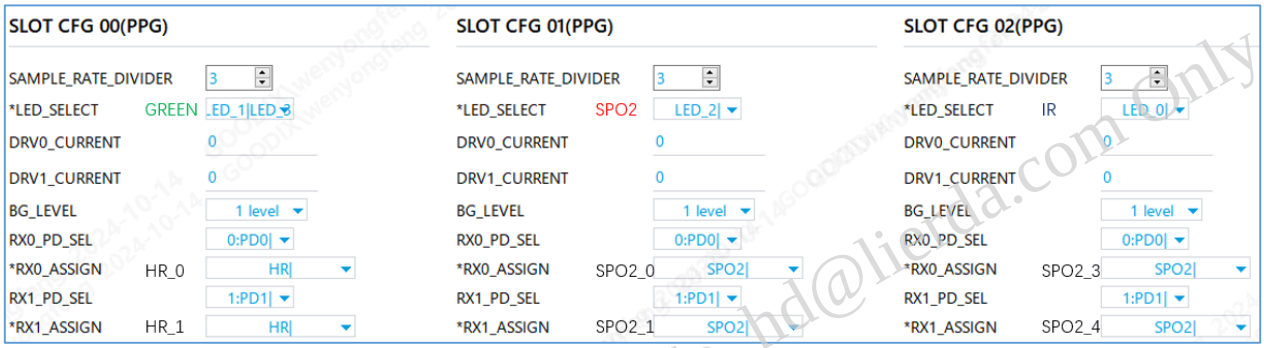

Figure 3-19 PD Assign

Set ALGO_GREEN_CH0 to 0 at “algo_green_chnl” on “HR-VIR” page, which means using HR_0 as the first green channel, and set ALGO_GREEN_CH1 to 1.

Note:

Select the channel with the largest CTR as ALGO_CH0

Figure 3-20 HR Algorithm Channel

Set ALGO_RED_CH0 to 0 at “algo_red_chnl” on “SPO2-VIR” page, which means using SPO2_0 as the first red channel, and set ALGO_IR_CH0 to 2 at “algo_ir_chnl”.

Note:

Select the channel with the largest CTR as ALGO_CH0

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Figure 3-21 SPO2 Algorithm Channel

3.9 Understanding Configuration Plot

After saving configuration file, the PC tool will generate a configuration plot. From the plot, we can find these parameters:

1. Function List
2. Slot sample rate
3. PD & LED map
4. Power and slot time

Notice that the PD and LED in the plot means the physic PD and physic LED.

Example as follows (At page plot), the sample rate of slot0 is 25Hz, the LED work at slot0 is LED1 and LED3, the PD work on RX0 at slot 0 is PD1, the raw data generated by RX0 at slot 0 is assigned to “HR” function, the power of slot0 is about 8ua, and the time of slot0 is about 112us.

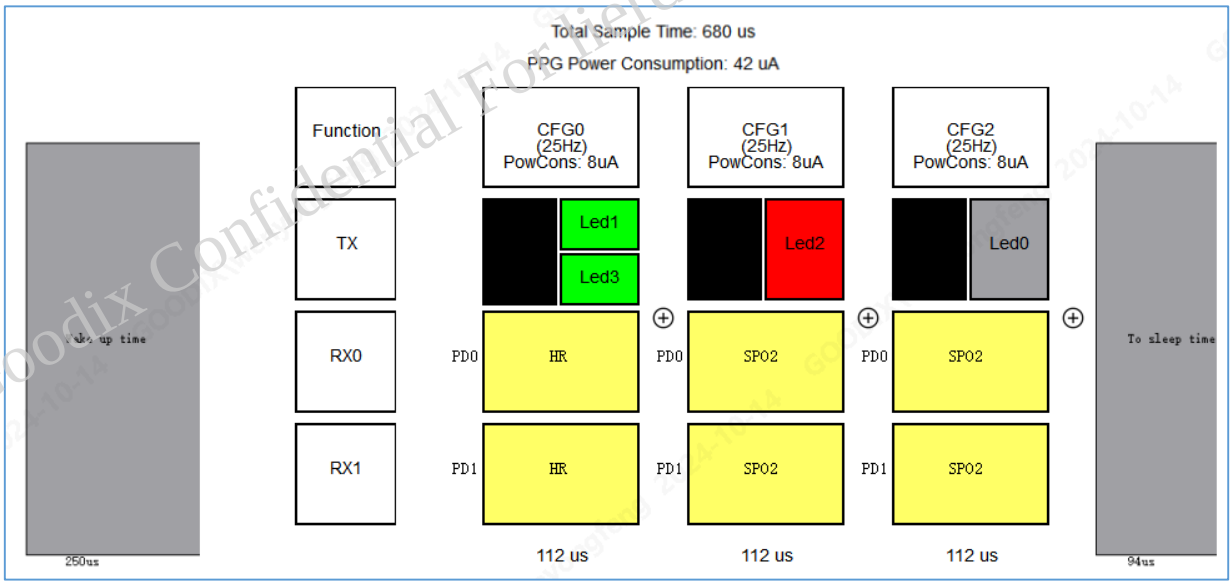

Figure 3-22 configuration plot

3.10 PPG graphical configuration

This chapter shows how to quickly configure PPG through a graphical interface.

3.10.1.1 Plot Configuration

3.10.1.1.1 Move Slot

Keep clicking on one slot, and then drag it to the another position.

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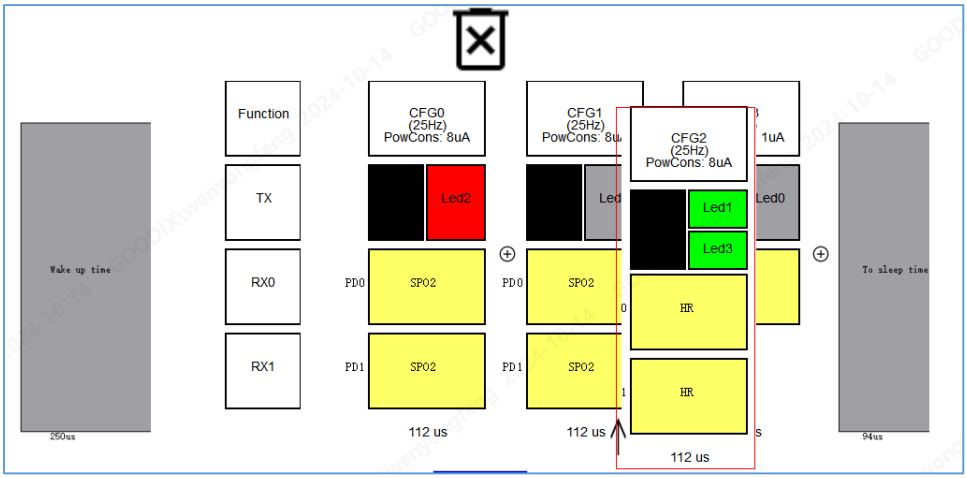

Figure 3-23 Move slot

3.10.1.1.2 Delete Slot

Click on one slot, and then drag it to the position of the icon above.

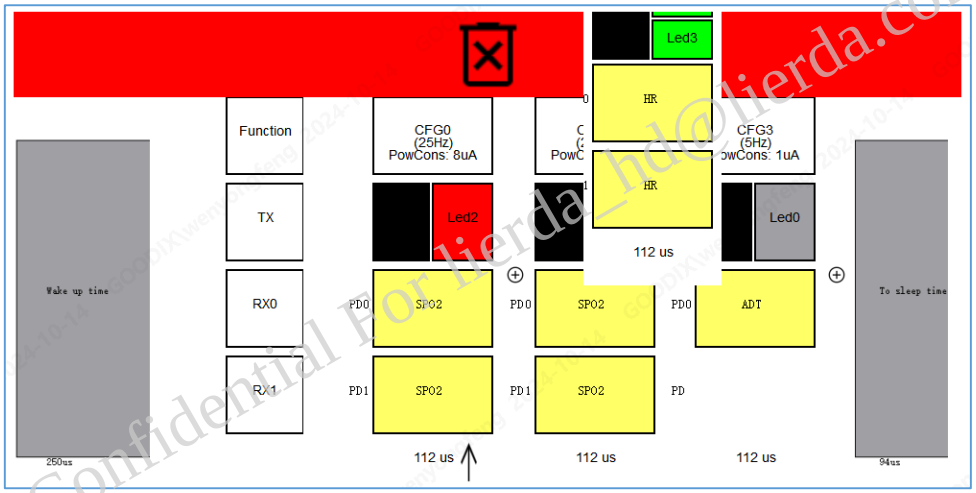

Figure 3-24 Delete slot

3.10.1.1.3 Insert New Slot

Click the icon to insert a new slot.

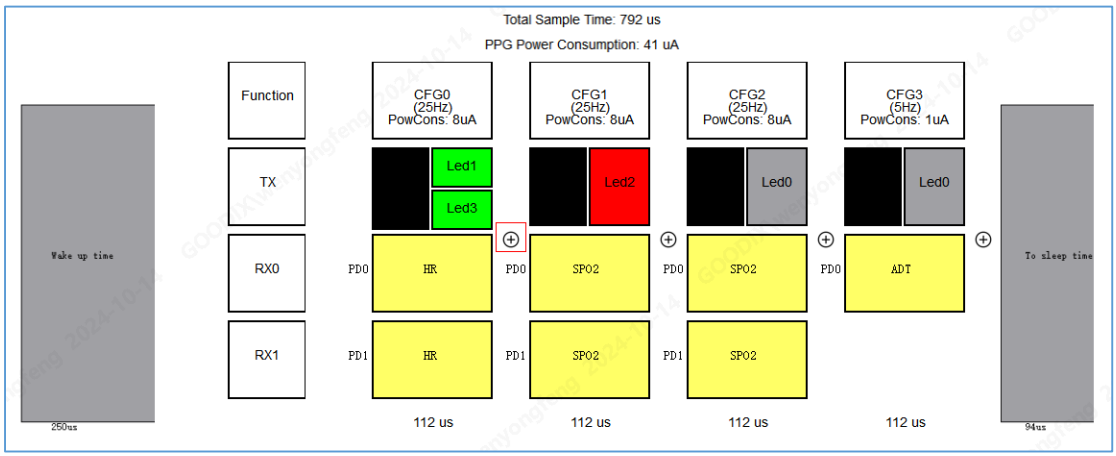

Figure 3-25 Insert new slot

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3.10.1.1.4 Copy Slot

Right-click and select "Copy this PPG slot to end".

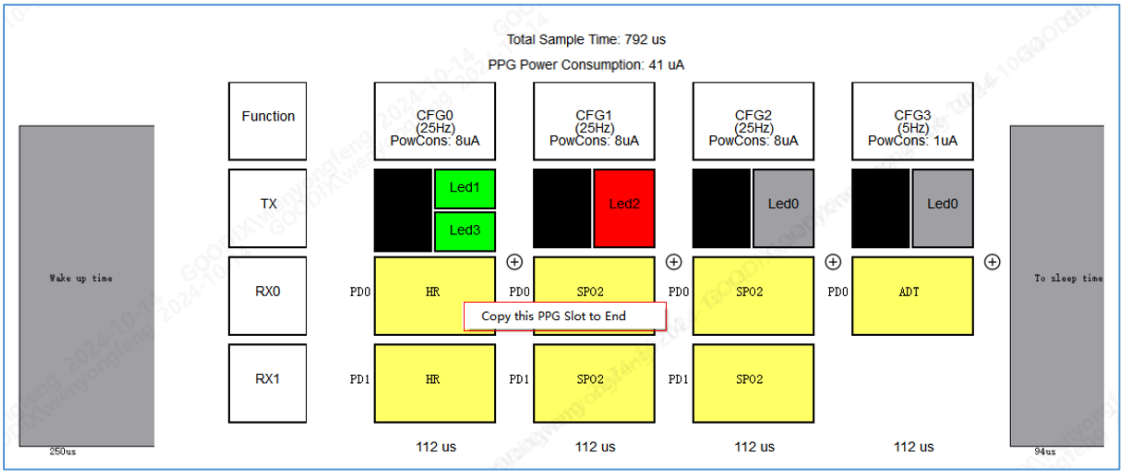

Figure 3-26 Copy slot

3.10.1.2 Slot configuration

Click one slot to enter Slot graphical configuration. The configuration items are the same with those at chapter 3.3 .

3.10.1.2.1 LED Configuration

Configure LED items at page “LED CONFIG”.

Notice:

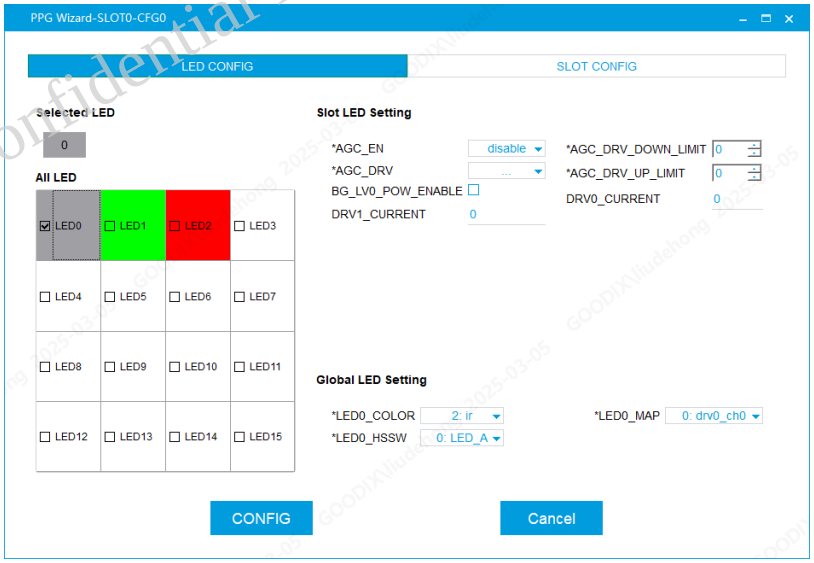

Figure 3-27 LED CONFIG

3.10.1.2.2 Slot Parameters

Configure other slot parameters at page “SLOT CONFIG”. Click the “RX” icons to switch selected “RX”.

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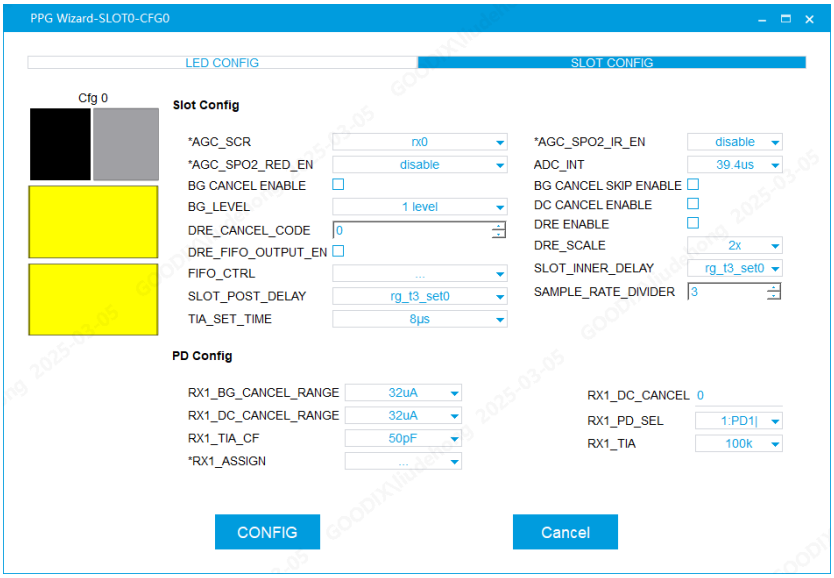

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4 Examples

4.1 Modify parameters

4.1.1 Add channels

4.1.1.1 Add PD channels

Set RX_PD_SEL to choose PDs and set RX_ASSIGN (If necessary).

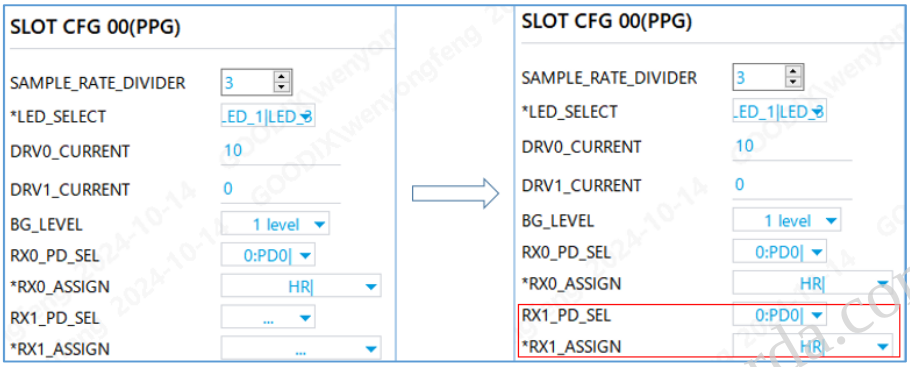

Figure 4-1 Add PD channels

4.1.1.2 Add a slot

Step 1. Set slot index

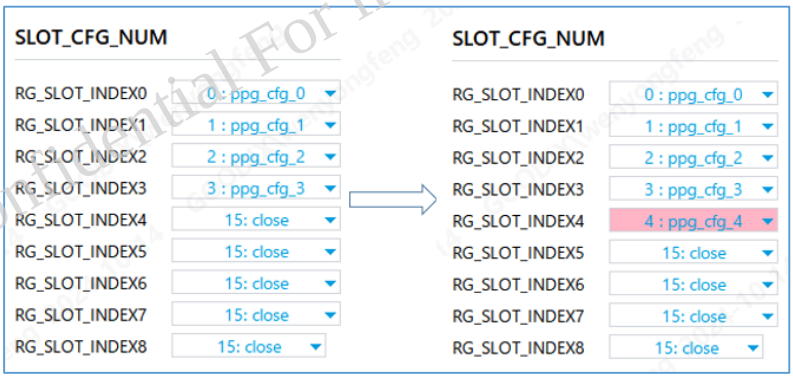

Figure 4-2 Set slot index

Step 2. Set slot parameters.

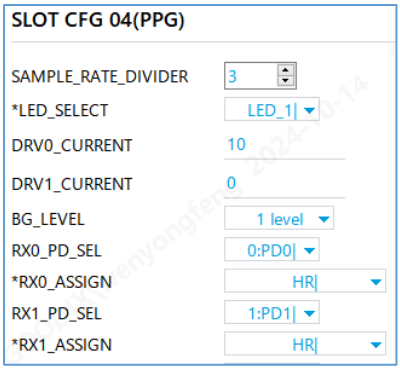

Figure 4-3 set slot parameters

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4.1.2 Add Functions

Step 1. Select new functions.

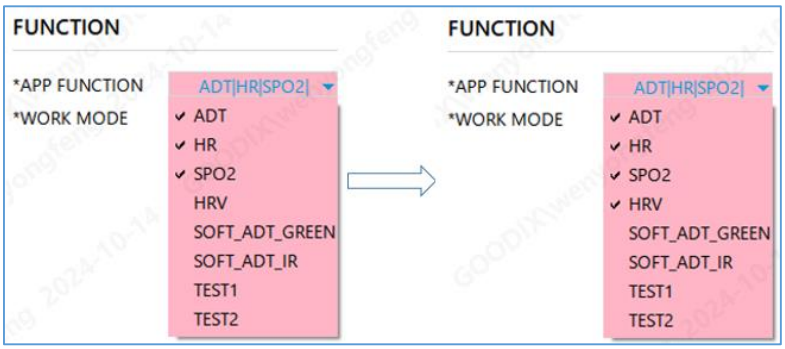

Figure 4-4 Select new functions

Step 2. Assign RX data to new functions.

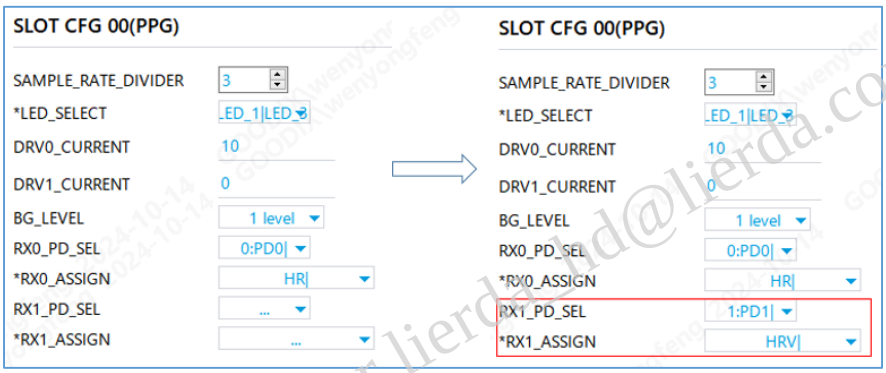

Figure 4-5 Set Rx Assign

Step 3. Set algorithm parameters.

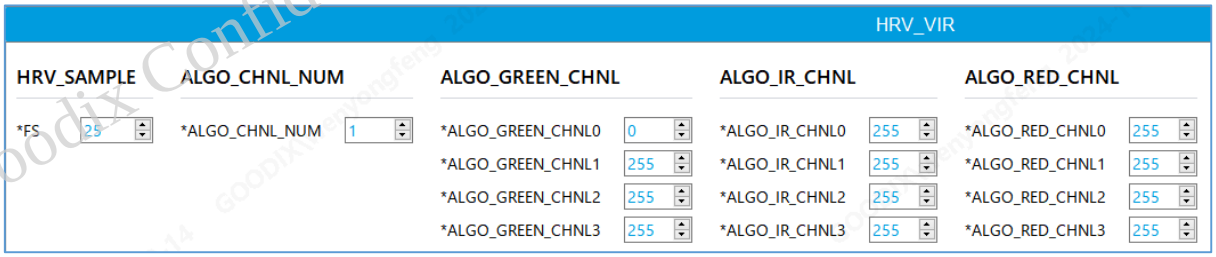

Figure 4-6 Algorithm parameters

4.2 Complete configuration

It is recommended to modify the parameters from the reference configurations. Please contact GOODIX to get the reference configuration files at first.

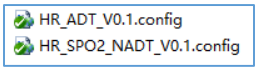

Figure 4-7 Reference configuration

Assume that there are some differences between GOODIX's optical path and user's optical path. And there are only 1 PD (PD0) connected to gh3036 on user's sensor board. The examples in this chapter will show how to modify the reference configuration to fix adapt new hardware.

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4.2.1 HR configuration

Step 0. Load reference configuration file “HR_ADT.config”. Browse information of this configuration file through the plot and we could find 2 slots have been set. Read chapter 3.9 to learn to get more information from plot.

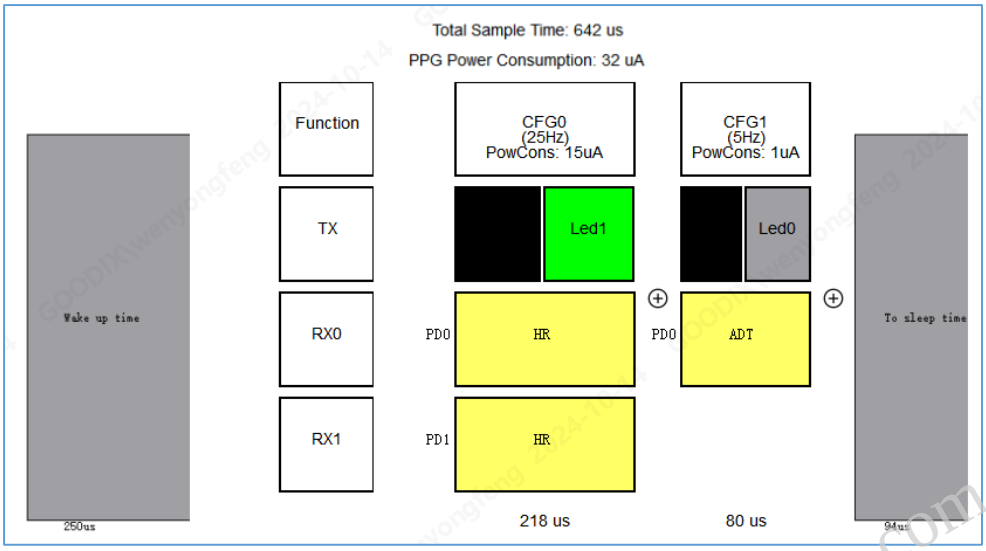

Figure 4-8 Reference configuration

Step 1. Select functions. The function list of this configuration includes ADT and HR.

Figure 4-9 Function list

Step 2. Configure LED&PD. The configuration rules are shown at chapter 3.2.1 and chapter 3.2.2 .

Step 3. Set slot parameters. Notice that there are only 2 PDs and the LED driver which drive green LED has been changed. Keep the other slot parameters and we can check if the parameters are correct according to appendix table.

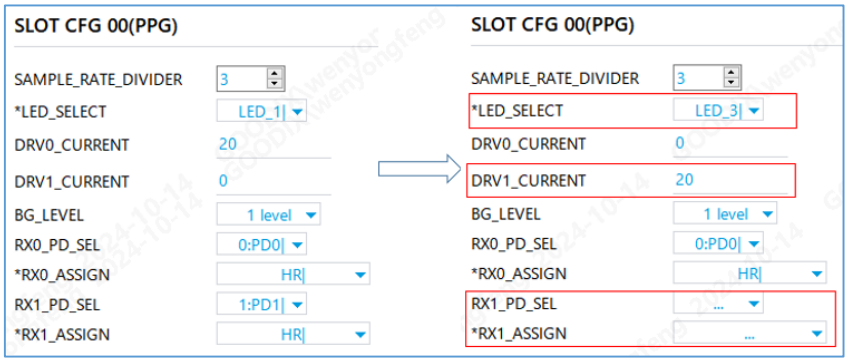

Figure 4-10 Set slot parameters

Step 4. Set agc parameters. Change LED driver and keep other parameters.

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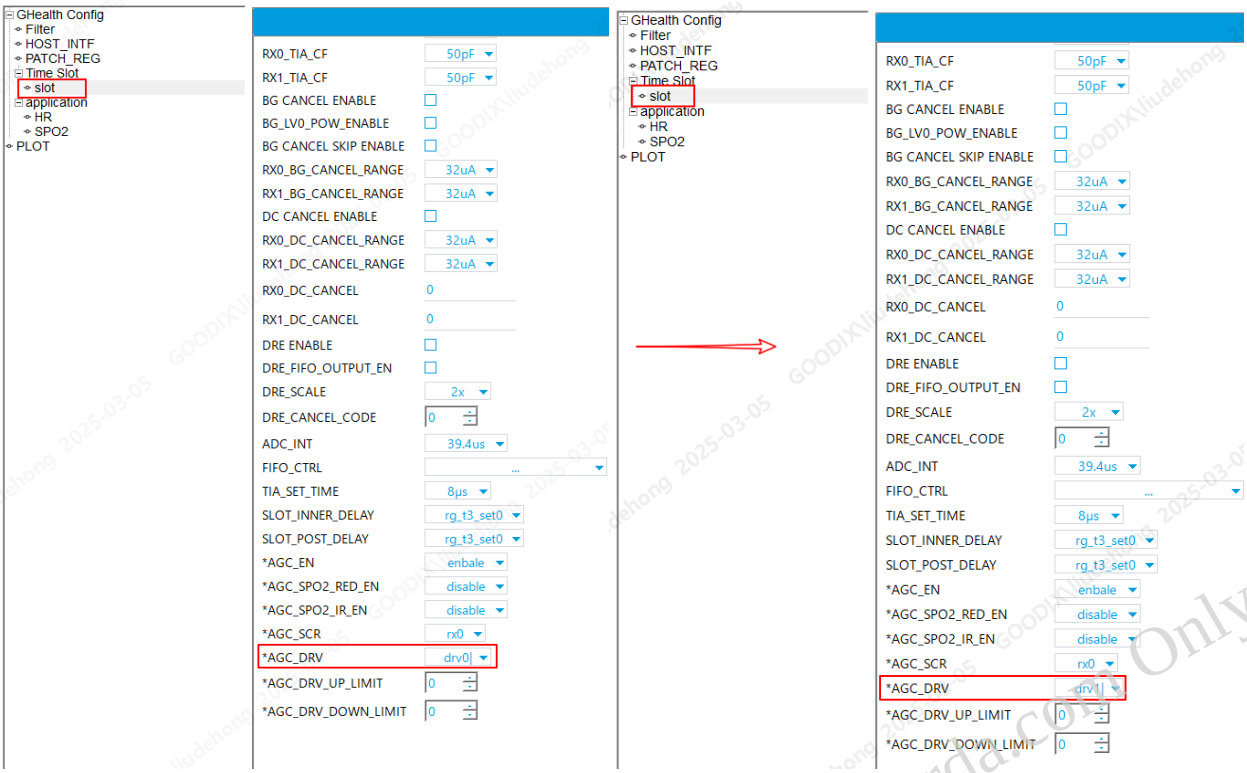

Figure 4-11 Set agc parameters

Step 5. Set algorithm parameters. Reduce 1 green channel because there are only 1 PD.

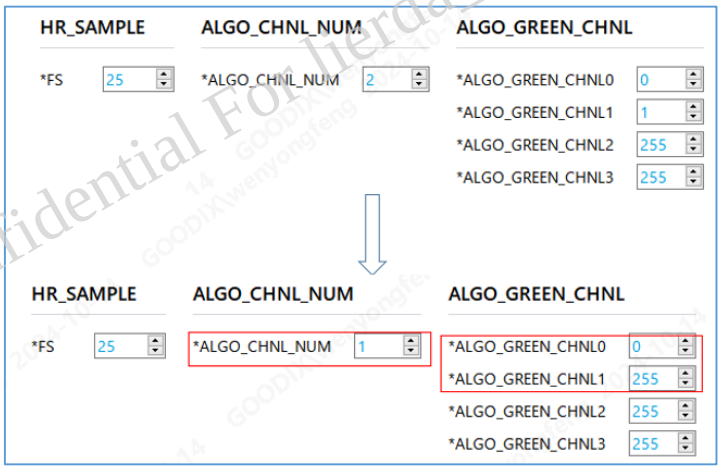

Figure 4-12 Set algorithm parameters

Step 6. Save configuration file and check if it is correct from the plot. And we could find there are only two PDs on this configuration.

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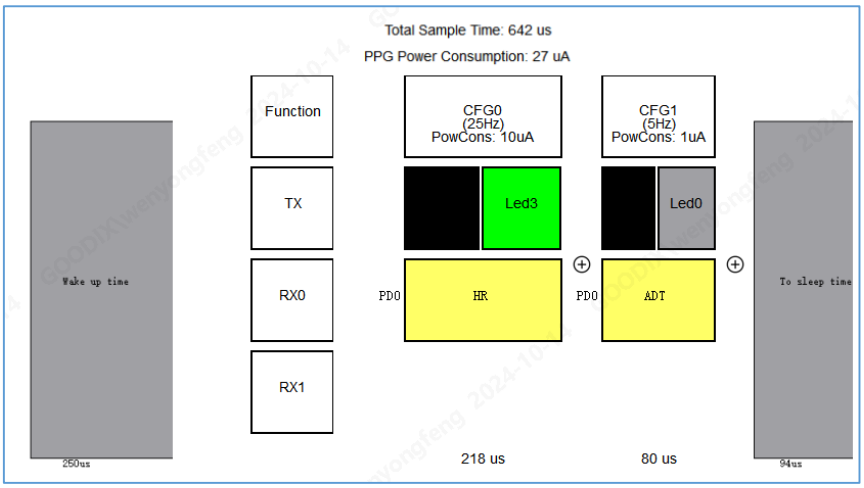

Figure 4-13 1pd HR configuration

4.2.2 Multifunction configuration

This example will add SPO2 function to a configuration which includes HR and ADT functions.

Step 0. Load reference configuration file.

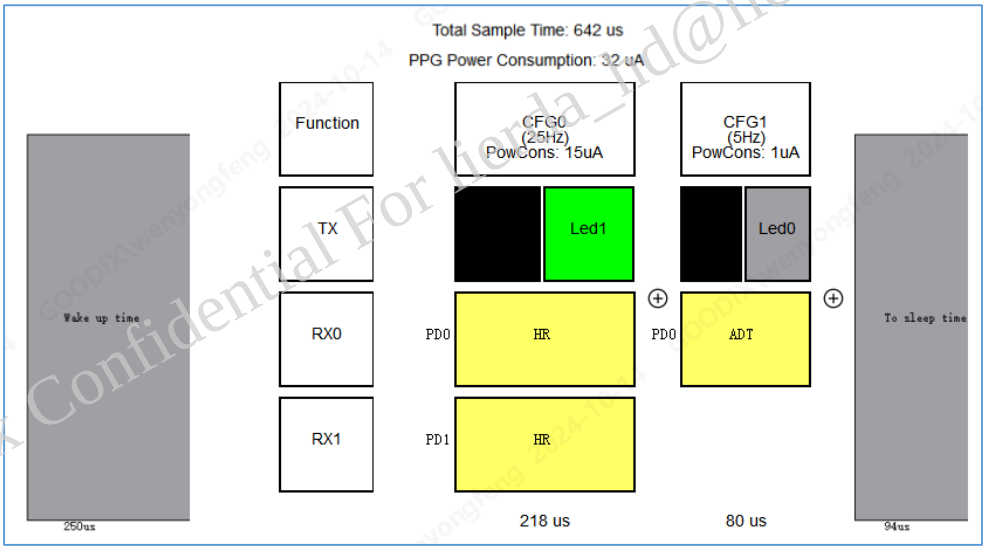

Figure 4-14 Reference configuration

Step 1. Add SPO2 at function list.

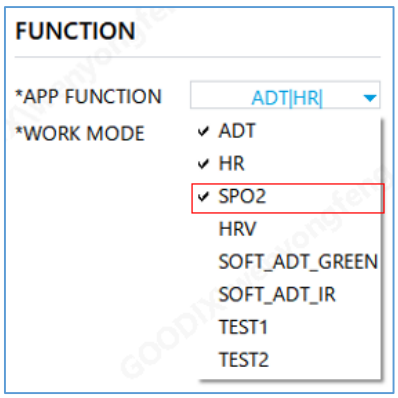

Figure 4-15 Function list

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Step 2. Set PD&LED. The process is the same as chapter 4.2.1 and it won't be repeated at this chapter again.

Step 3. Set slot parameters.

Add 2 slots for SPO2 and move the ADT slot to the last position.

Figure 4-16 Add SPO2 slots

Select LEDs and PDs for each slot, and assign the channels to SPO2. The other slot parameter can refer to the appendix table.

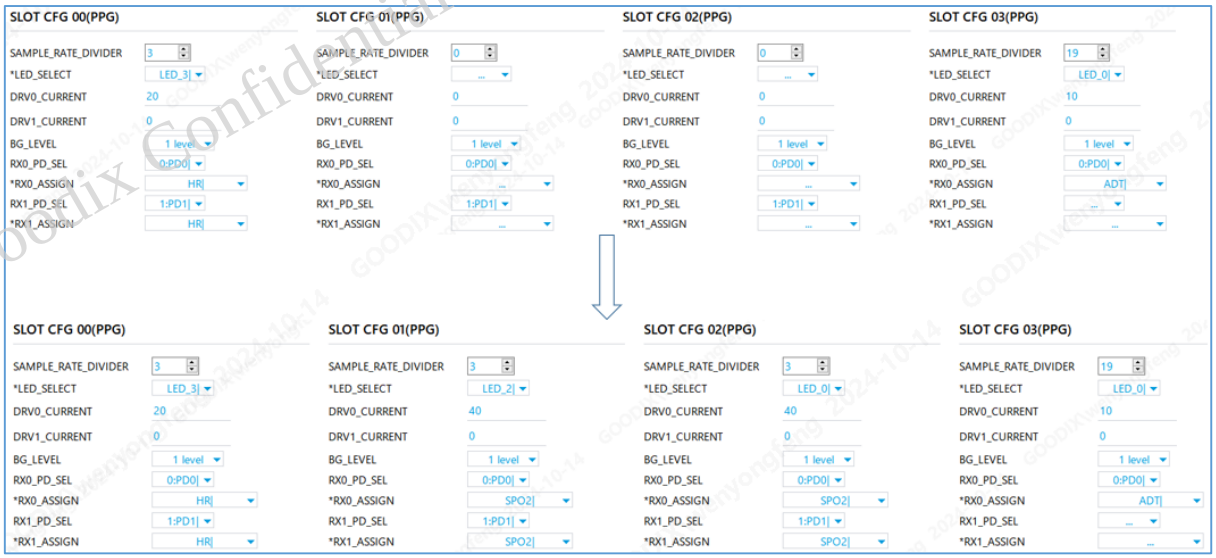

Figure 4-17 Set slot parameters

Step 4. Set Algorithm channel.

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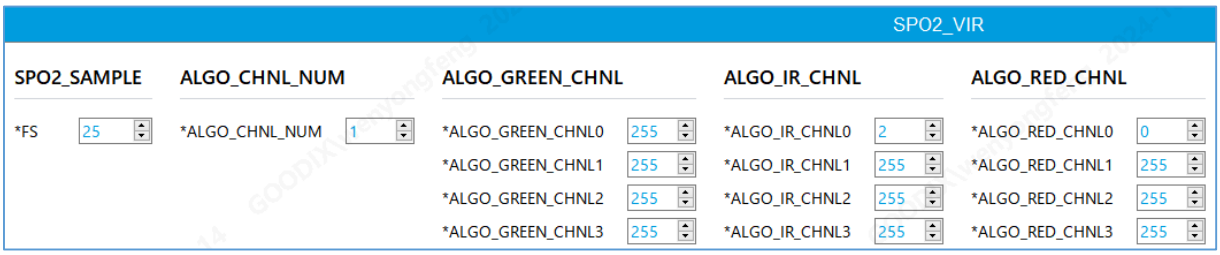

Figure 4-18 Set algorithm parameters

Step 5. Save configuration file.

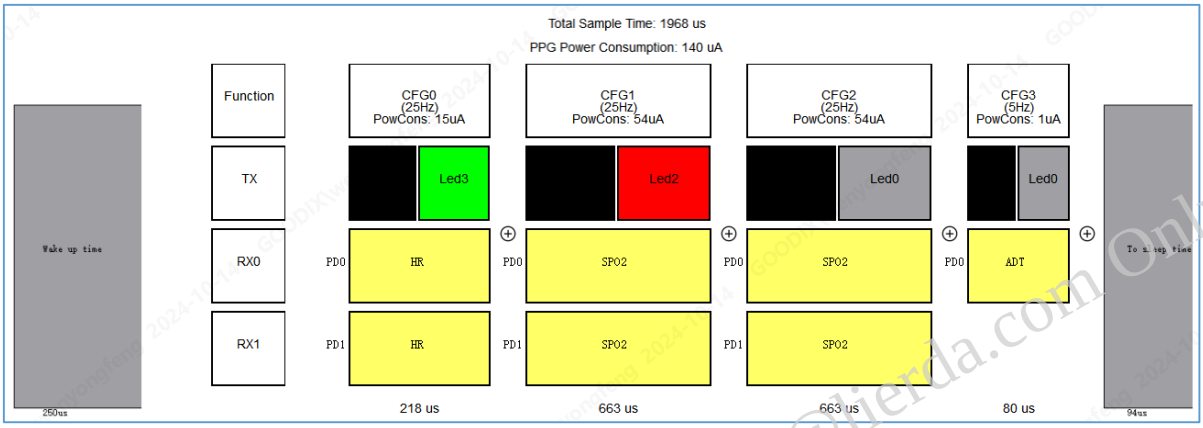

Figure 4-19 SPO2 added configuration

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4.2.3 HR DRE configuration

Step 0. Load “HR_DRE_ADT.config”. CFG0 is HR function, and CFG1 is HR function with DRE.

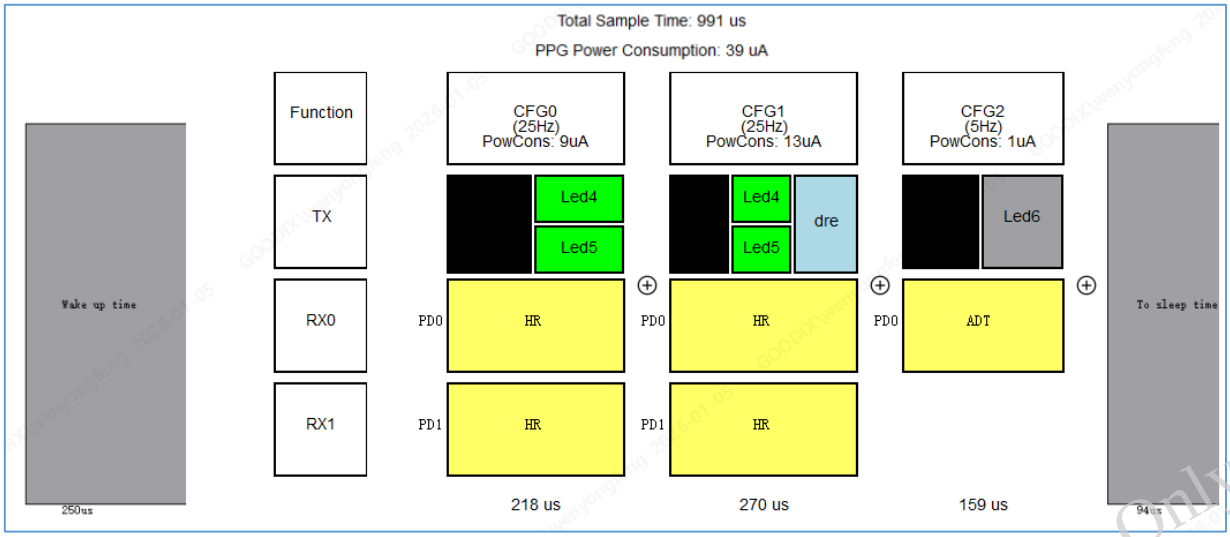

Figure 4-20 HR DRE Reference configuration

Step 1. Set DRE parameters in “slot”. “DC CANCEL ENABLE” and “DRE ENABLE” must be enable when using DRE. Set “DRE_CANCNEL_CODE” only in HR function, and the larger the code value, the faster data approaches zero after DRE is triggered. Set “DRE_FIFO_OUTPUT_EN” if use DRE AGC in GH3036 SDK. And other description of DRE registers are shown at chapter 3.3.4.

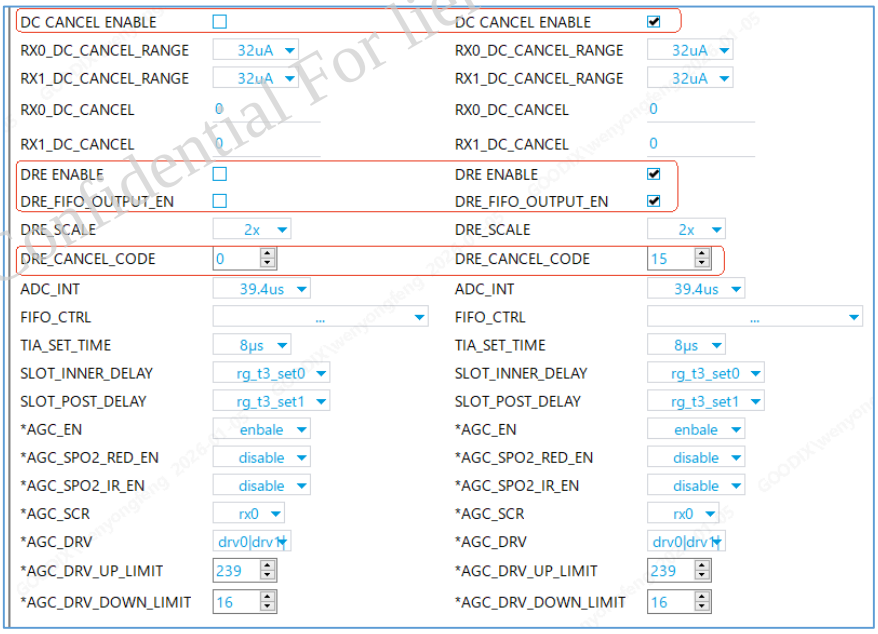

Figure 4-21 HR DRE Reference configuration

Step 2. Set DRE parameters in “SAMPLE”. Set “RG_DRE_DC_THR”, a larger value makes DRE less likely to trigger and increases the probability of saturation; a small value makes DRE easier to trigger, which incurs power consumption. It is recommended that do not change “RG_DRE_EN_THR_CNT”, “DRE SAT HANDLE ENABLE” and “RG_DRE_CODE_MODE”. The description of DRE registers are shown at chapter 3.6.2.

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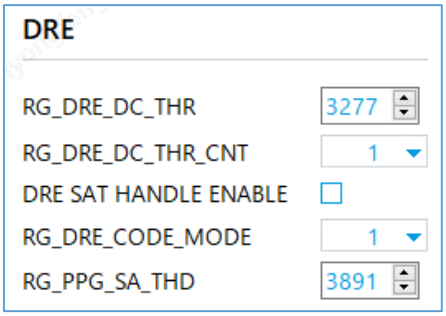

Figure 4-22 HR DRE Reference configuration

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5 Appendix

GH3036
customer project

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6 VLED Low-Voltage Power Supply Configuration

When using Vled low-voltage power supply, the SDK parameters must be configured accurately to ensure proper algorithm performance, including the software AGC parameters and the wear-detection algorithm parameters.

6.1 software AGC configuration

When using the VLED low-voltage power supply scheme (to eliminate the external boost converter), the software AGC module must achieve accurate control of the LED current. In the SDK file 'gh_hal_config.h', the macro definition GH_HAL_FINE_MAX_DRV_MA should be configured to the maximum drive current supported by a single DRV in the low-voltage application:

When one LED is driven by a single DRV, GH_HAL_FINE_MAX_DRV_MA = I_LED_MAX.

When two DRVs are connected in parallel to drive a single LED, GH_HAL_FINE_MAX_DRV_MA = I_LED_MAX / 2.

6.2 nADT configuration

Since the maximum LED current is limited under low-voltage conditions, the PPG signal amplitude becomes smaller when worn by users with dark skin. To prevent false "wear-off" detections in such cases, a parameter in the wear-detection algorithm needs to be reduced. However, lowering this parameter will degrade the performance of object-removal detection. The threshold modification procedure is as follows: In the SDK source file 'goodix_nadt_config.c', assign the value 500 to the structure member g_gh_nadt_cfg.sig_scale_thr.

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